Sign Language Transcription (SLT)

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Approach:

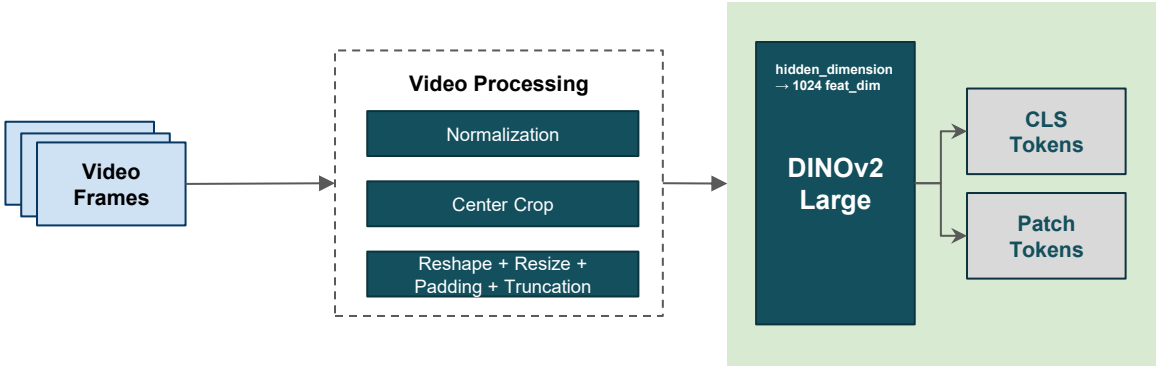
Gloss-Free Sign Language Recognition and Translation → Sign Language Transcription (SLT)

- ➔ Visual Features Extraction via Vision Transformer (DINOv2-Large)
- ➔ Transcription in English using Language Model

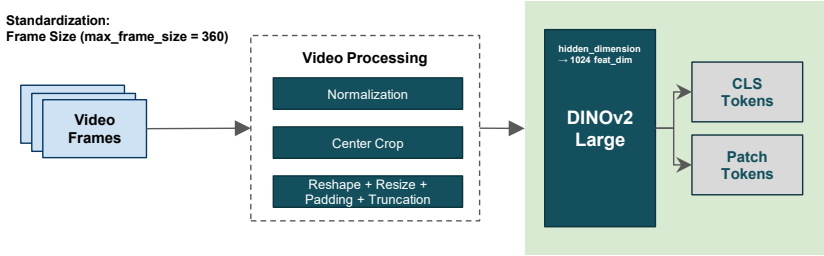
1. Overview: Dataset and Visual Features Extraction using DINOv2-Large

How2sign

→ How2Sign dataset - **Duarte et al. (2021)** Presented at **CVPR 2021**
→ Continuous American Sign Language (ASL) dataset
→ Corpus of more than 80 hours of sign language videos with corresponding English transcriptions
→ Consists of Instructional content across various domains



2. Processing of Visual Features with DINOv2-Large



Video Processing Pipeline:

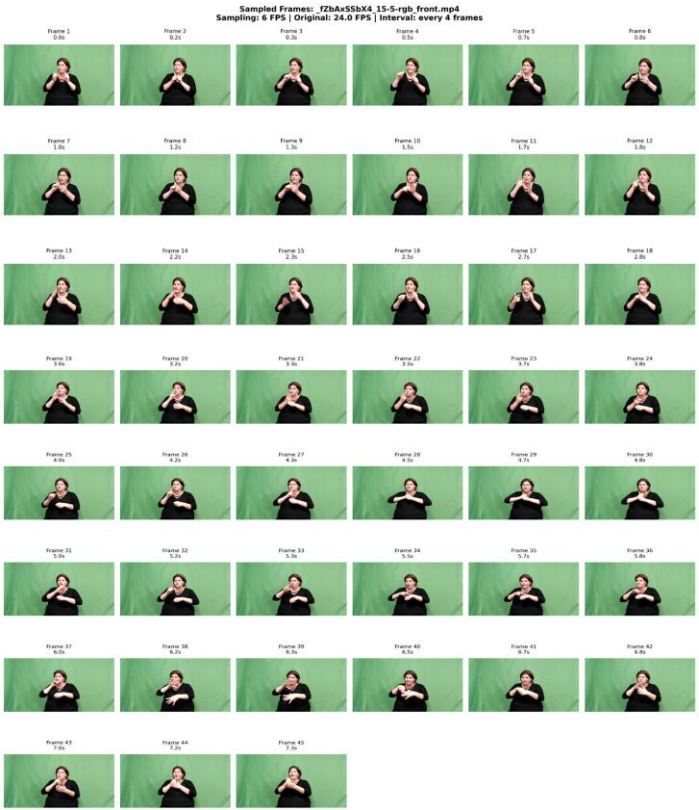
Frame Extraction → 6 frames per second (6 FPS) from given video of 24 FPS
Transformations → Extracted Frames Resize to 256×256, Center crop to 254×254, Random horizontal flip (±10°), Convert to tensors Standardize → 360 frames per given video (Duration ~ 1 minute video clips)

DINOv2-Large Model → Parameters: 304M, Patch size: 14, Embedding dimension: 1024

Feature Extraction:
Output feature shape per video:

- **CLS tokens: [360,1024] with 360 frames and a 1024-dim cls token per frame**
- **Patch tokens: [360, 256, 1024] with 360 frames, 256 patches per frame and 1024-dim feature vector per patch**

Frames Extraction Example from a Video



Corresponding Target Sentence:

“There's obviously a tension and the rhythm is in that, so it will help you develop that part.”

Approaches for Text Generation (Transcription)

➔ Processing of Extracted Visual Features (DINOv2-Large) with:

1. **Baseline Model:** Conditional Generation using BART Encoder-Decoder Model (**BART-139M**)
2. **Customized:** Causal (Prefix) Generation using SmolLM2 Decoder Model (**SmolLM2-360M**)

3. Processing of Textual Features with Language Models

Given: Target Sentence from Dataset → `Sentence`

(Baseline) BART-139M:

Tokenizer

- **Vocabulary Size: 50265**
- **BOS (Beginning of Sequence) token: `<s>` — ID: 0**
- **EOS (End of Sequence) token: `</s>` — ID: 2**
- **PAD (Padding) token: `<pad>` — ID: 1**
- **Results in: `[<BOS>Sentence<EOS>]`**
- **Padding Side = Right**
- **Results in: `[<BOS>Sentence<EOS>]`**

Training Objective - Conditional Generation:

- **Condition on extracted visual features to generate token embeddings from target sentence**
- **`[Condition: Visual_Features] → [Embedded_Text_Tokens]`**

SmolLM-360M:

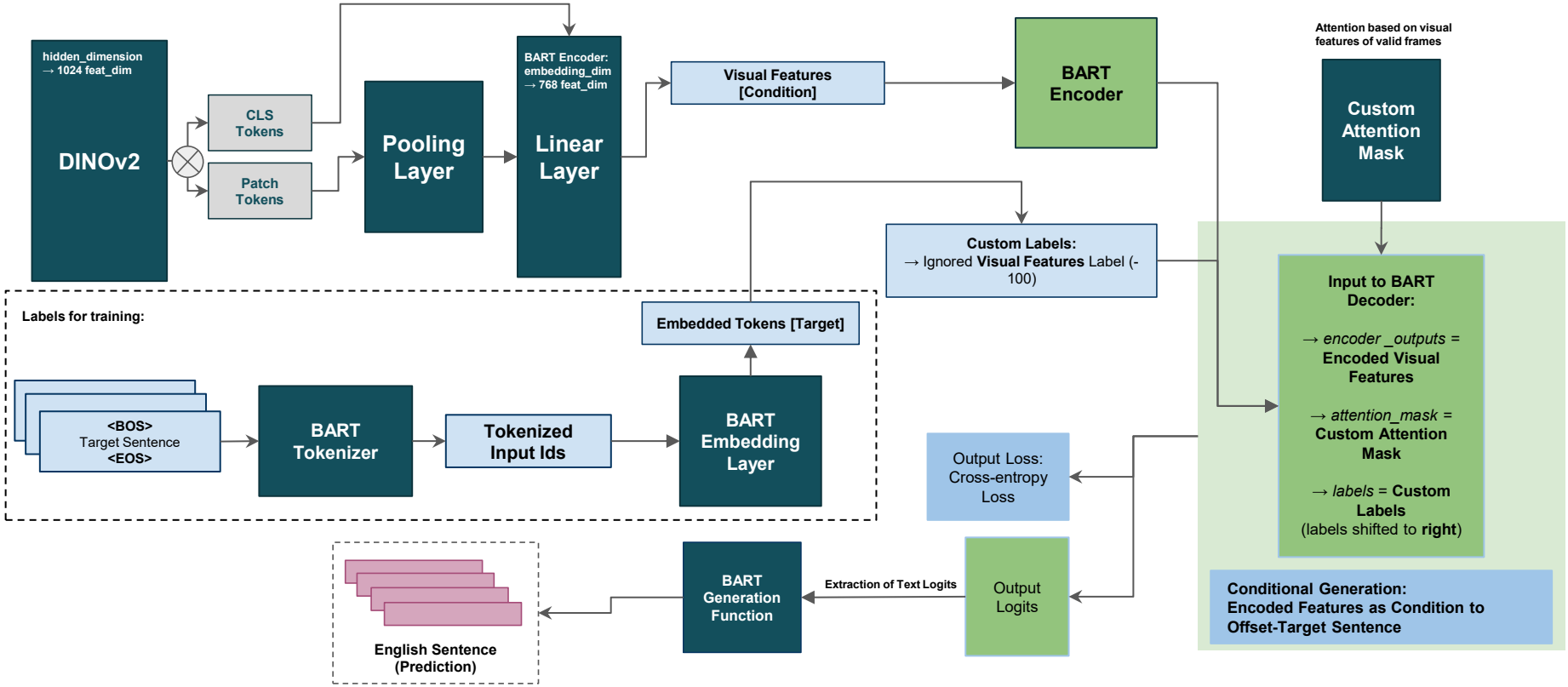
Tokenizer:

- Vocabulary Size: 49152
- **BOS token == EOS token**
- EOS token: **<|endoftext|>** ID: 0 (Default) sdfs
- **PAD** token not set up by default
- Added:
  - BOS token: **<BOS>** ID: 49152
  - PAD token: **<PAD>** ID: 49153
- New Vocabulary Size: 49154 (2 added tokens)
- Padding Side = Right
- Results in: **`[<BOS>Sentence<EOS>]`**

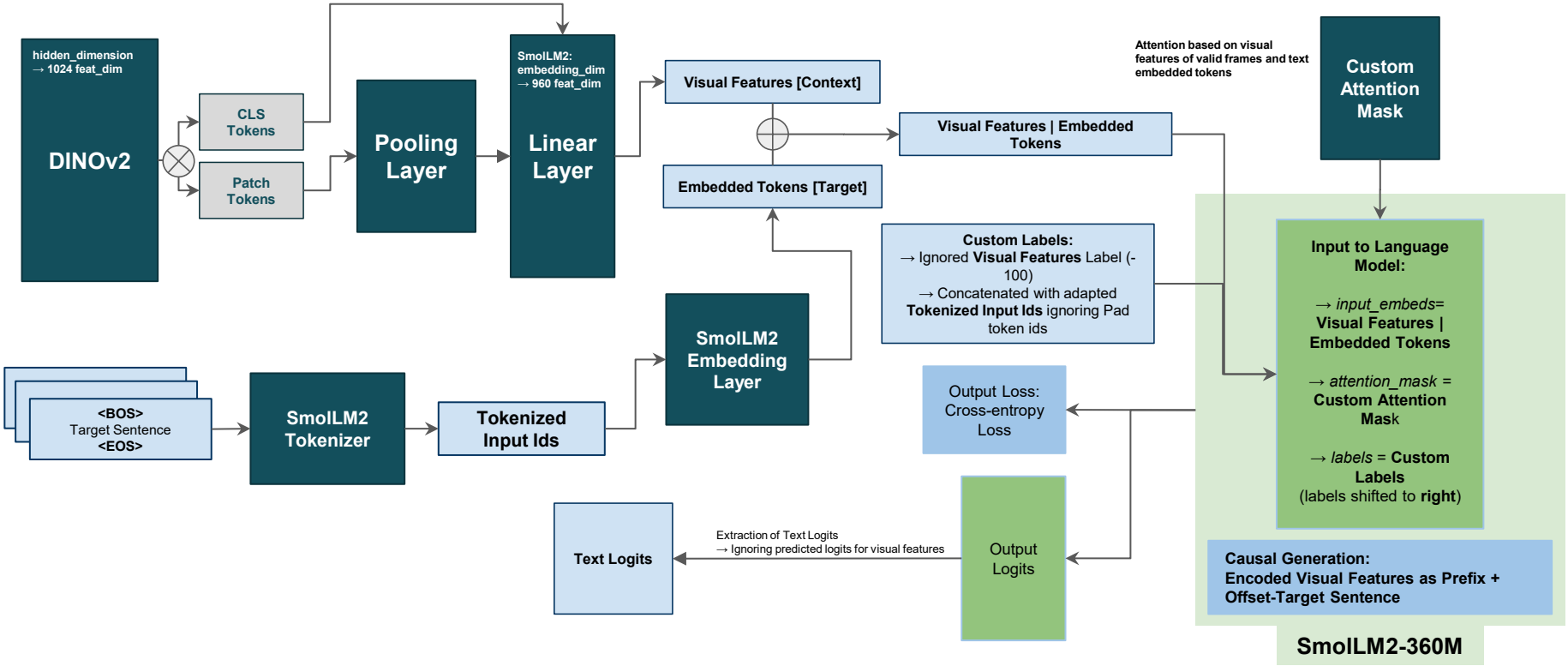
Training Objective - Causal Generation

- Concatenation of projected visual features with token embeddings from target sentence as prefix
- **`[Prefix: Visual_Features | Embedded_Tokens]`**

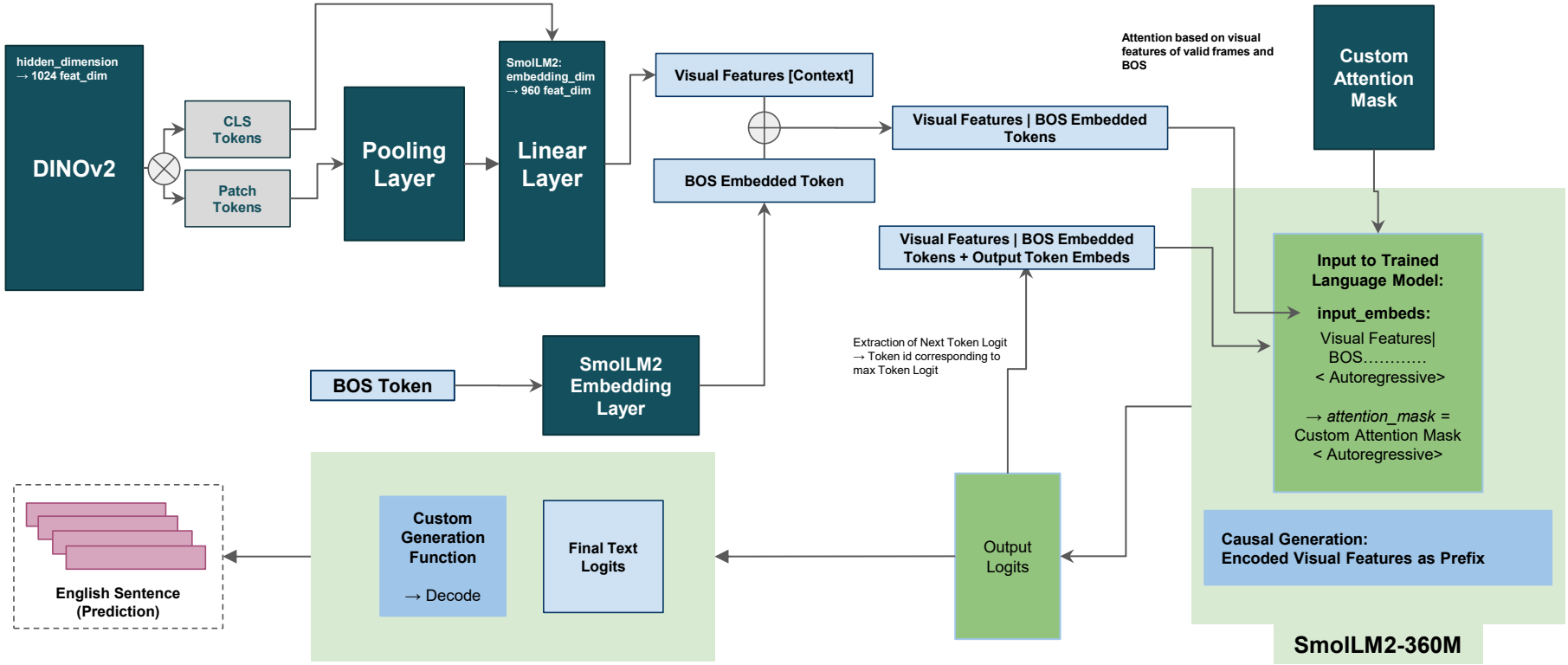
4. (Baseline) DINOv2 with BART-139M: Training and Inference Pipeline for SLT Model



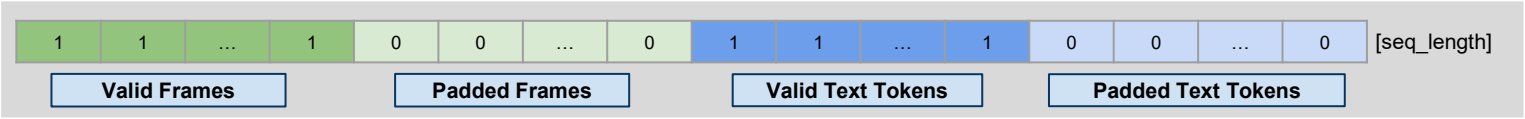
5. DINOv2 with SmolLM2-360M: Training Pipeline for SLT Model



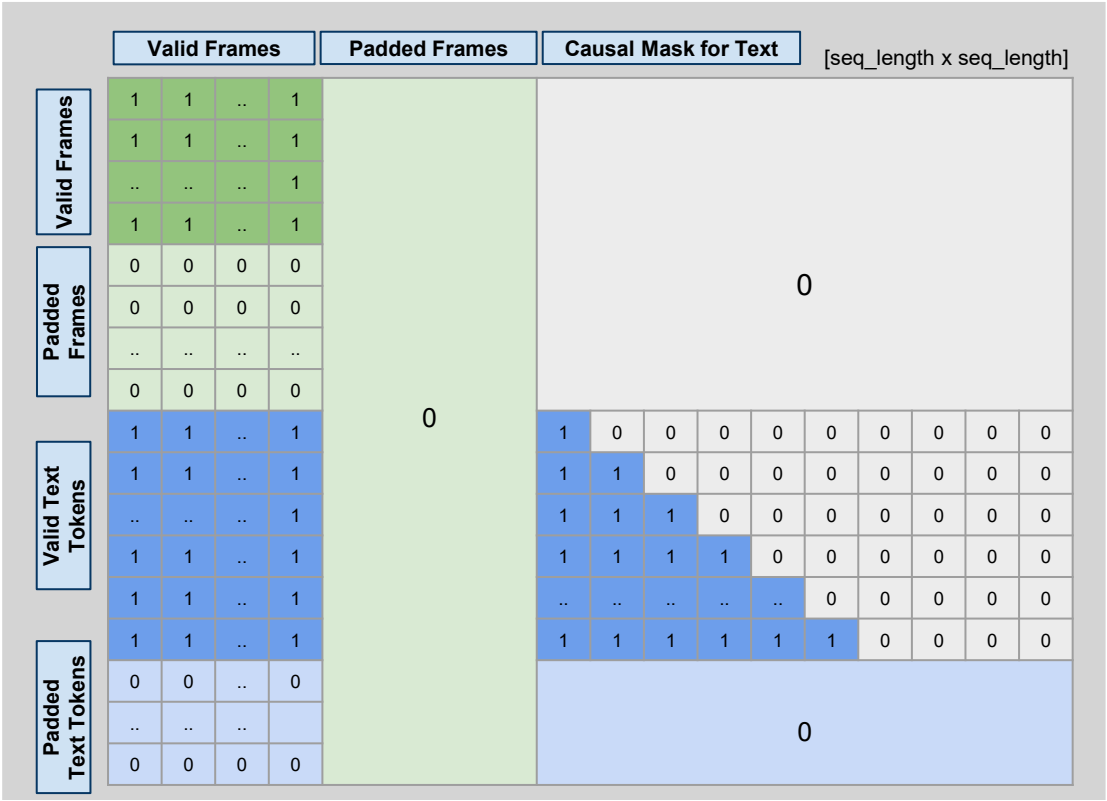
6. DINOv2 with SmolLM2-360M: Inference Pipeline with SLT Model



7. Custom Attention Mask for Visual Features and Textual Features



- Valid visual tokens attend to each other
- Padded visual tokens do not interfere
- Valid text tokens attend to all valid visual tokens
- Valid text tokens attend causally to prior text tokens
- Padded text tokens do not interfere



8. Evaluation Metrics

1. **Cross-Entropy (Token level accuracy): Loss function**

2. **Perplexity (Token level accuracy): Exponential of average cross-entropy loss. Lower score indicates more confidence and accuracy**
   → ~10-20: reasonable
   → 50: poor generalization/ out of distribution

3. **BLEU-4 Score (Local fluency): Measures 4-gram precision between predicted and reference sequences. Standard for benchmarking translation and generation tasks**
   → ≥0.5: high-quality

4. **ROUGE-L Score (Global meaning): Measures longest common subsequence between predicted and reference text to capture global content similarity.**
   → ≥0.5: strong content overlap